Lesson 3 — Regression

Technologies for Artificial Intelligence

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Before we start

- Exercise 2 was due today
- The pipeline you built is the one we fit models into from now on

Today: the first real model

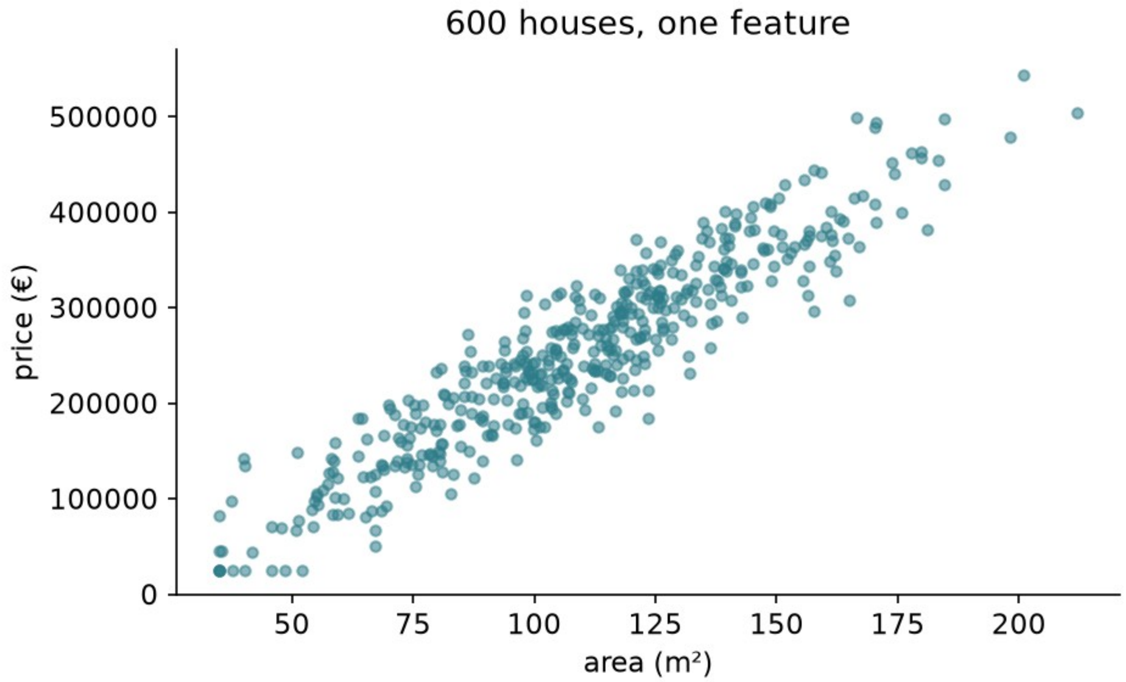
- The model and what it claims
- The exact solution — and why it sometimes fails
- Gradient descent
- Curves, and the price of flexibility
- Ridge and Lasso

One dataset, all lesson

600 houses. Six measurements. A price.

And we know the coefficients that generated it.

What we are trying to predict



What a linear model claims

Each feature contributes a fixed amount per unit, and the contributions add up.

- Every square metre: the same 2,400 €
- Every kilometre out: the same −6,500 €

Fitting means minimising something

We need one number saying how badly a candidate model is doing.

Adding up the errors fails: +50,000 and −50,000 cancel to zero.

Why squared, not absolute

- **Differentiable everywhere** — no corner at zero
- **Punishes one large error more** than several small ones
- Under Gaussian noise, it **is** maximum likelihood

The cost function — mean squared error

$$J(w,b) = \frac{1}{2m}\sum_{i=1}^{m}\left(\hat{y}^{(i)} - y^{(i)}\right)^2$$

Trying $\hat{y} = 2{,}400 \cdot \text{area} + 45{,}000$

Area (m^2)	True (€)	Predicted (€)	Error $\hat{y} - y$	Error^2
80	240,000	237,000	−3,000	9.0×10^6
120	320,000	333,000	+13,000	1.69×10^8
200	540,000	525,000	−15,000	2.25×10^8

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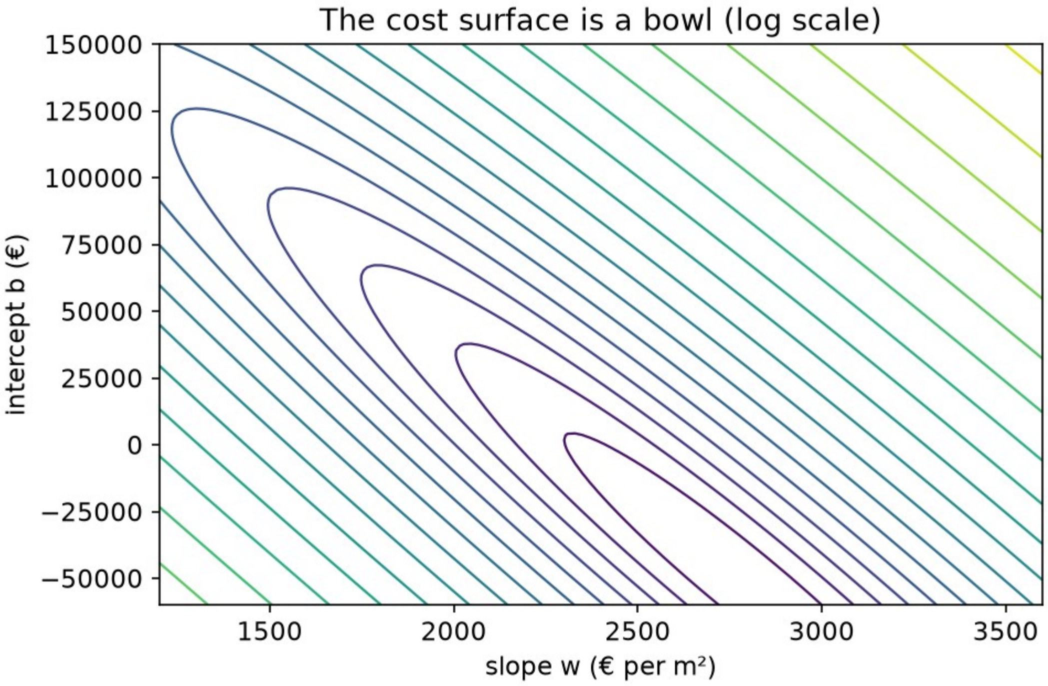
Squared error spends its attention on the worst house

- The 3,000 € error contributes about **2%** of the total
- The two large errors contribute the other **98%**
- Halving the 15,000 € error cuts the cost by **42%**

The cost is a bowl



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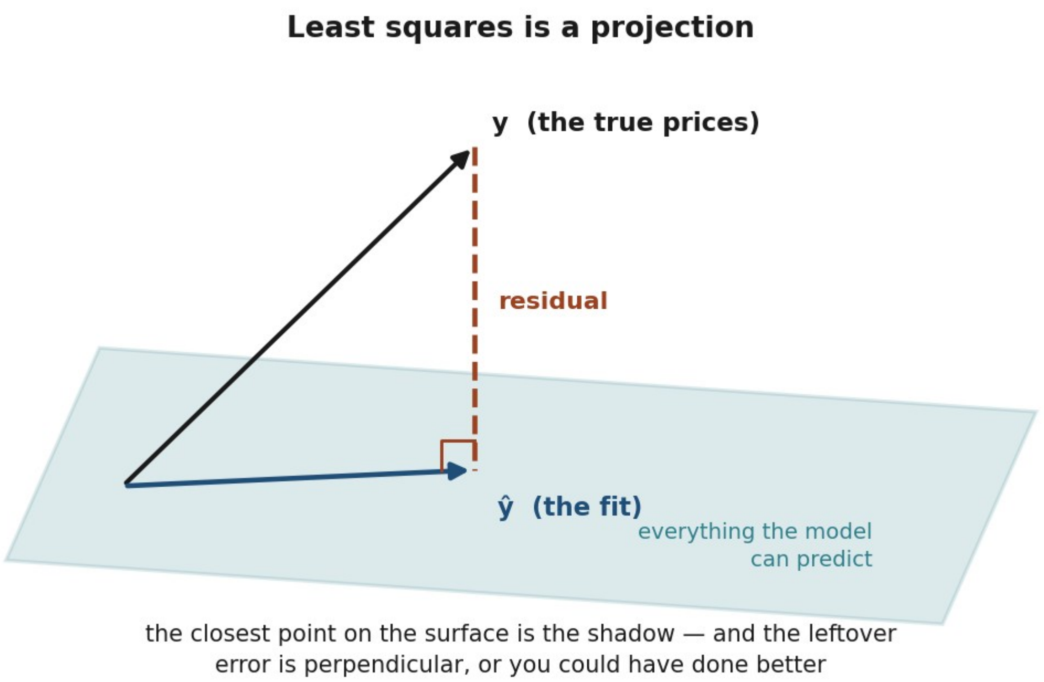
The picture behind the exact solution



The normal equation

$$\theta = (X^\top X)^{-1} X^\top y$$

The same equation, by hand

Quantity	Value
the three houses	80, 120, 200 m^2 → 240k, 320k, 540k
$X^T X$	rows (3, 400) and (400, 60,800)
$X^T y$	(1,100 , 165,600)
determinant	22,400
slope w	**2,536 €/m^2** — the truth is 2,400
intercept b	28,600 €

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And it really is the minimum

- The second derivative is $X^T X$ — never negative in any direction
- So the cost is **convex**: one bottom, no local traps
- Any stationary point is *the* answer, not *an* answer

So why learn anything else?

- $(X^T X)^{-1}$ costs about n^3 operations
- Two columns carrying one fact → **no inverse exists**
- Nearly-redundant columns → an inverse that is useless

How stretched is the valley?

The **condition number** compares the steepest direction with the shallowest.

- Housing features as recorded: **285**
- The same features, standardised: **3.4**
- Add `area_sqft` beside `area_sqm`: **2,286**

The picture behind gradient descent

You are on a hillside in fog.

You cannot see the valley — but you can feel which way the ground slopes.

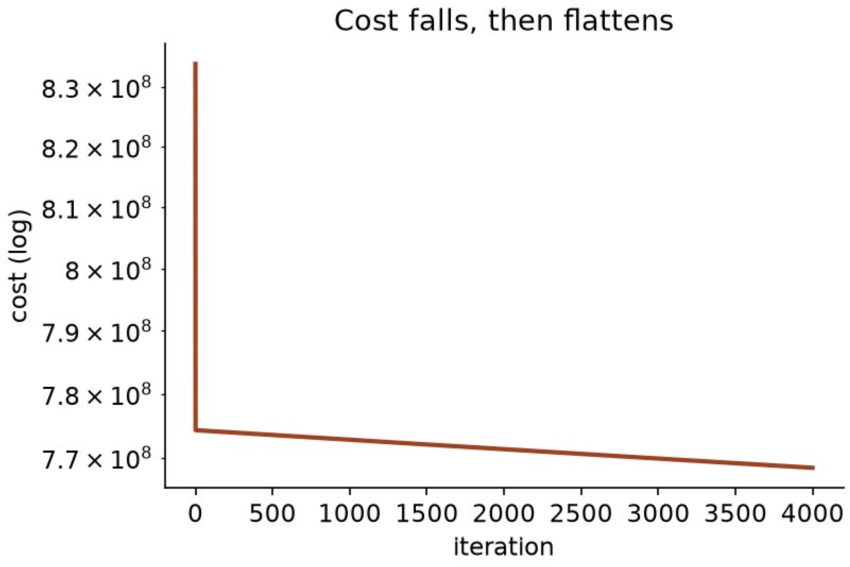
The update rule

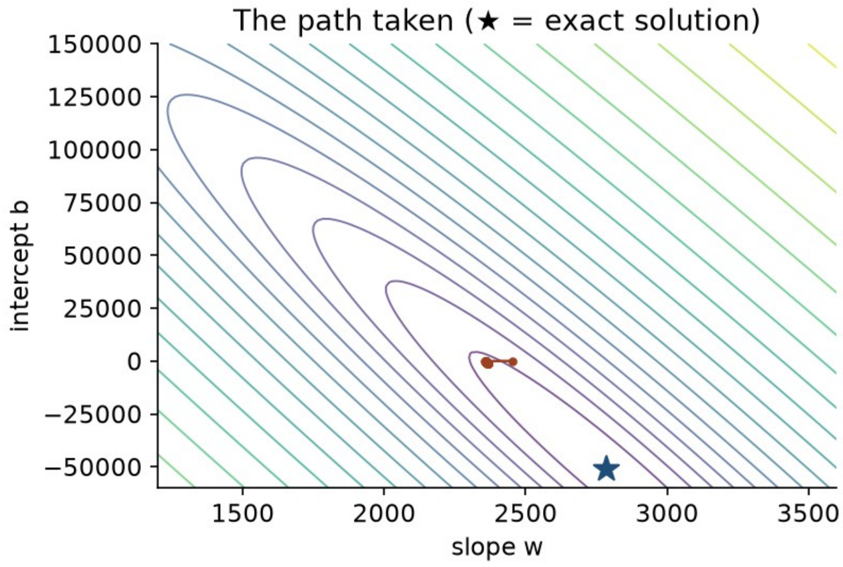
$$w \leftarrow w - \alpha \frac{\partial J}{\partial w}, \qquad \frac{\partial J}{\partial w_j} = \frac{1}{m}\sum_i \left(\hat{y}^{(i)} - y^{(i)}\right) x_j^{(i)}$$

One step, from w = 0 and b = 0

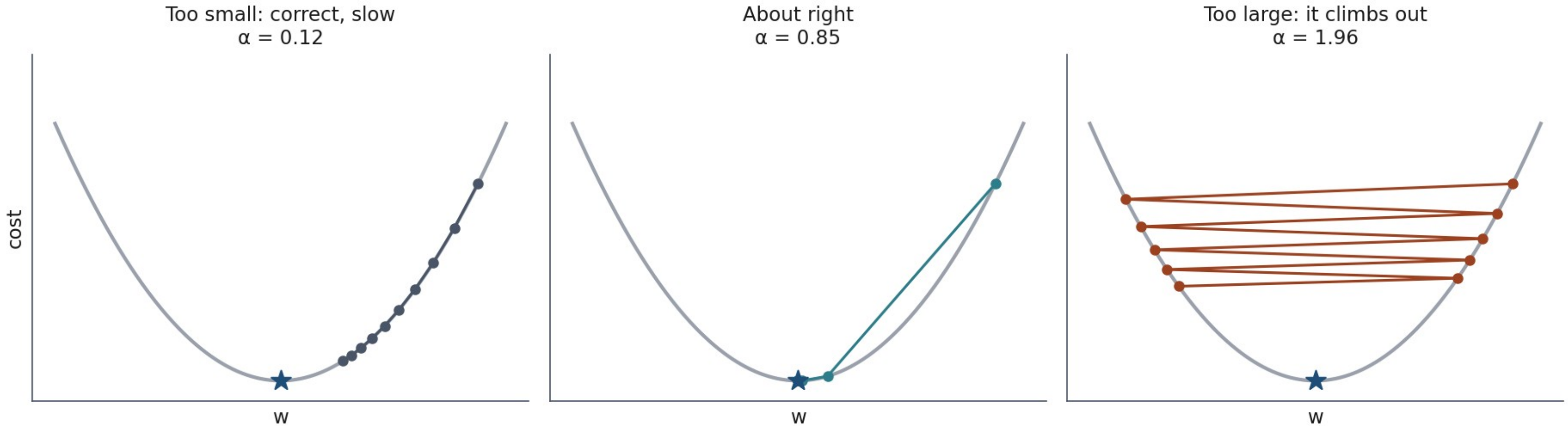
$\alpha = 10^{-5}$	Gradient	After one step
slope w	−63,600	0.636
intercept b	−390	0.0039

The path it takes





Three learning rates



Notebook 1, live

The model, the cost, the exact solution, the iterative one — from scratch.

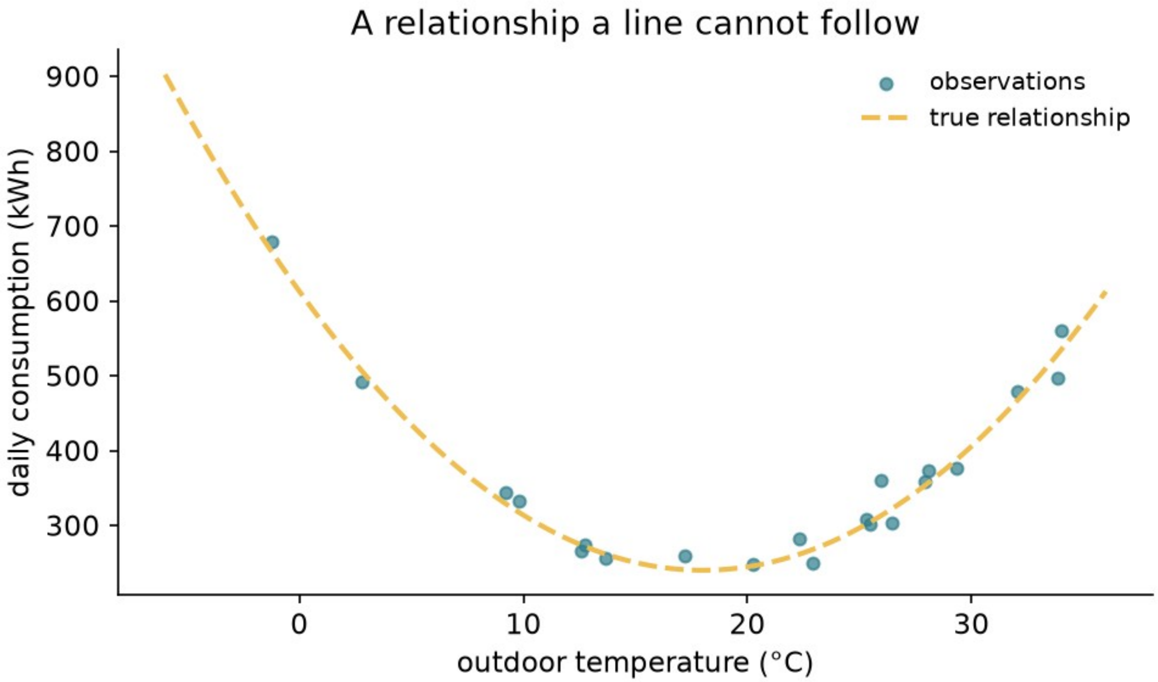
Break

What if the relationship bends?

Energy consumption against temperature: heating in the cold, cooling in the heat.

No straight line follows that.

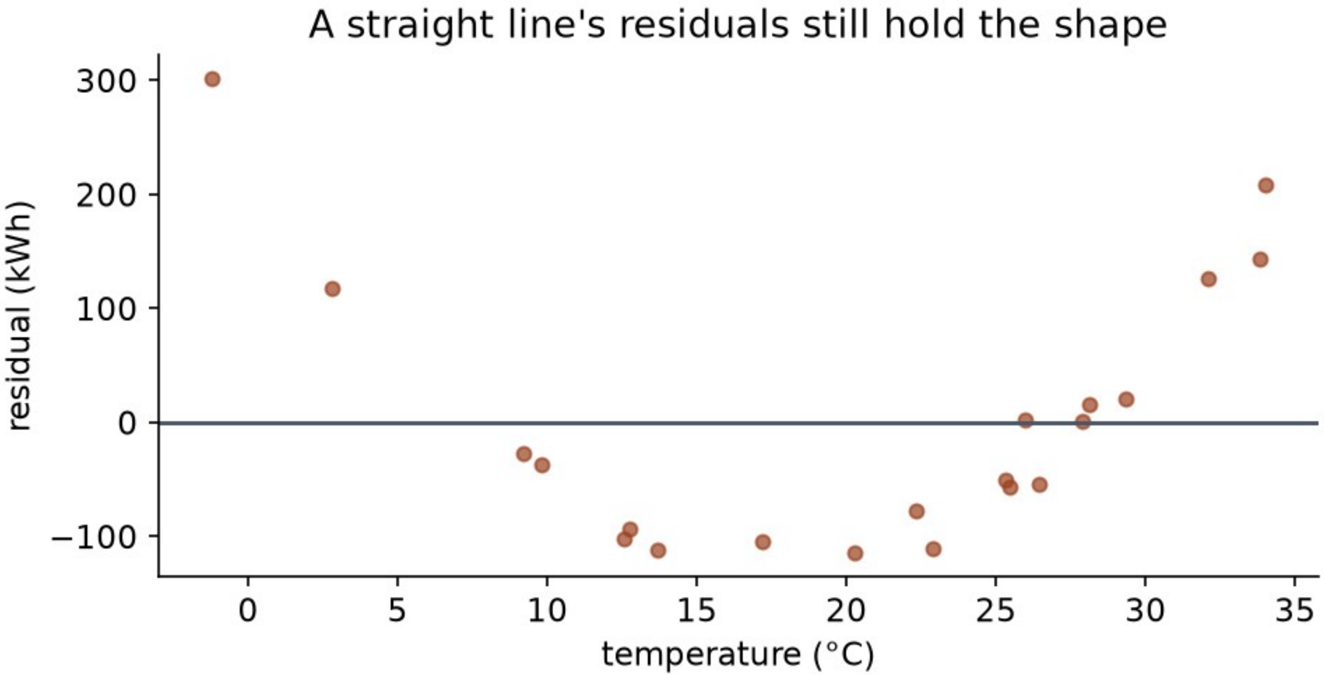
A relationship a line cannot follow



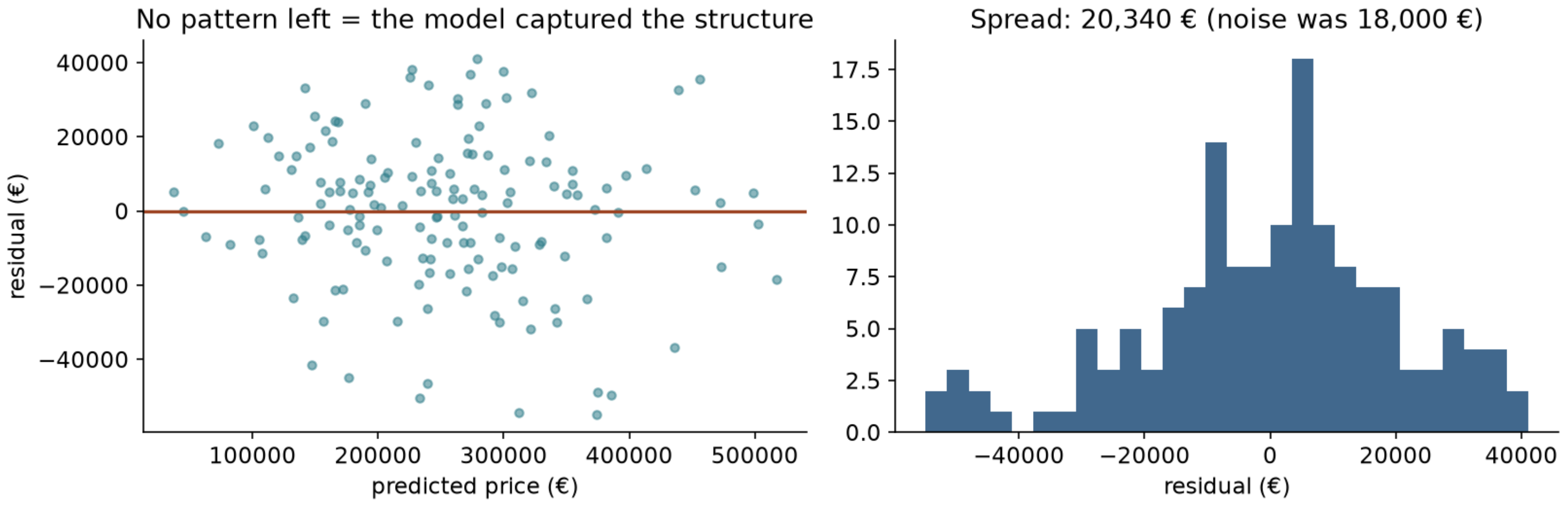
“Linear” means linear in the coefficients

$$\hat{y} = w_1 t + w_2 t^2 + b$$

The straight line's residuals keep the shape



What “nothing left” looks like



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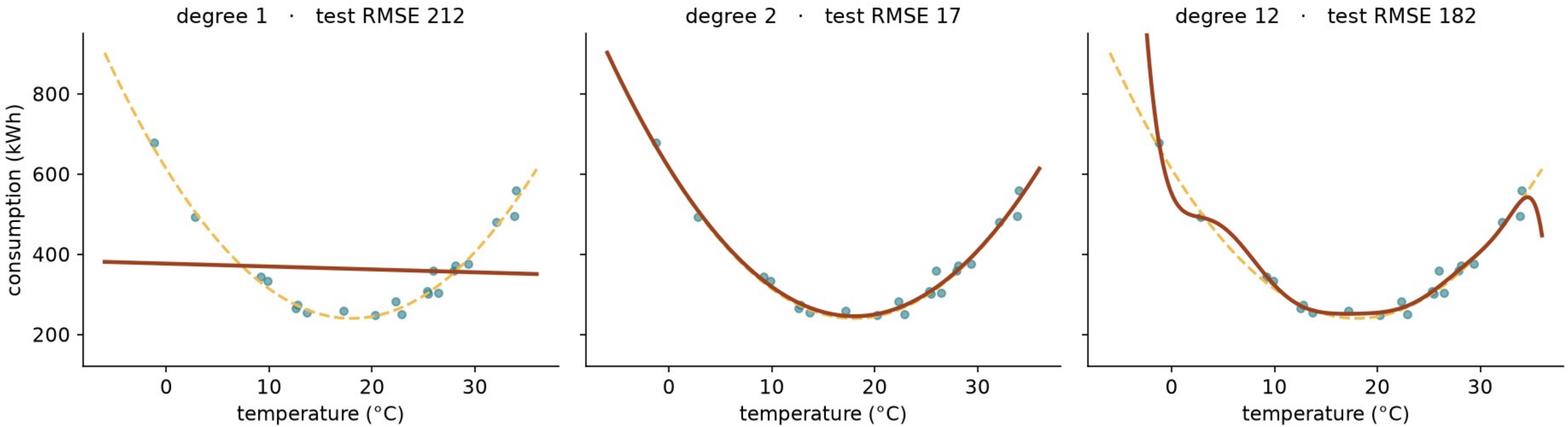
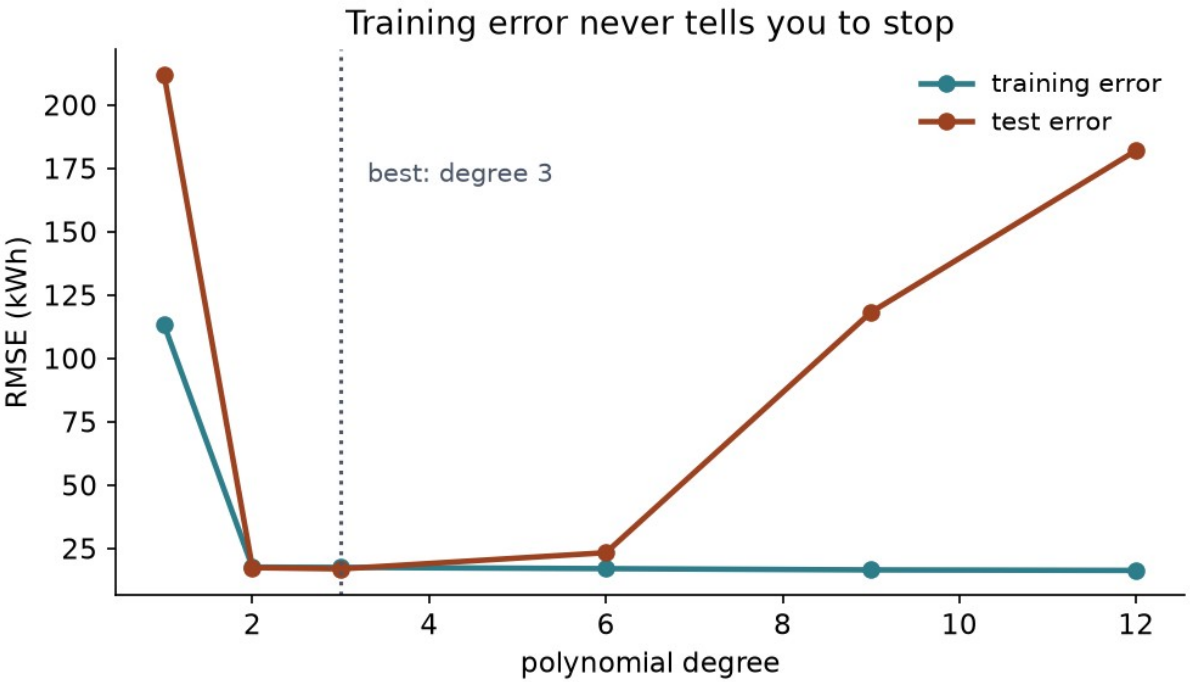
So use more flexibility?



Root mean squared error (RMSE), by degree

Degree	Train RMSE	Test RMSE
1	113.3	212.1
3	17.5	**16.9**
9	16.5	118.2
12	16.3	182.0

The gap is the overfitting



Notebook 2, live

Fit the curve, push the degree up, and watch the test error turn.

Why flexibility goes wrong

Largest coefficient, degree 2: **411**

Largest coefficient, degree 12: **247,514**

Charge for size

$$J_{\text{ridge}} = \text{MSE} + \lambda \sum_j w_j^2 \qquad J_{\text{lasso}} = \text{MSE} + \lambda \sum_j |w_j|$$

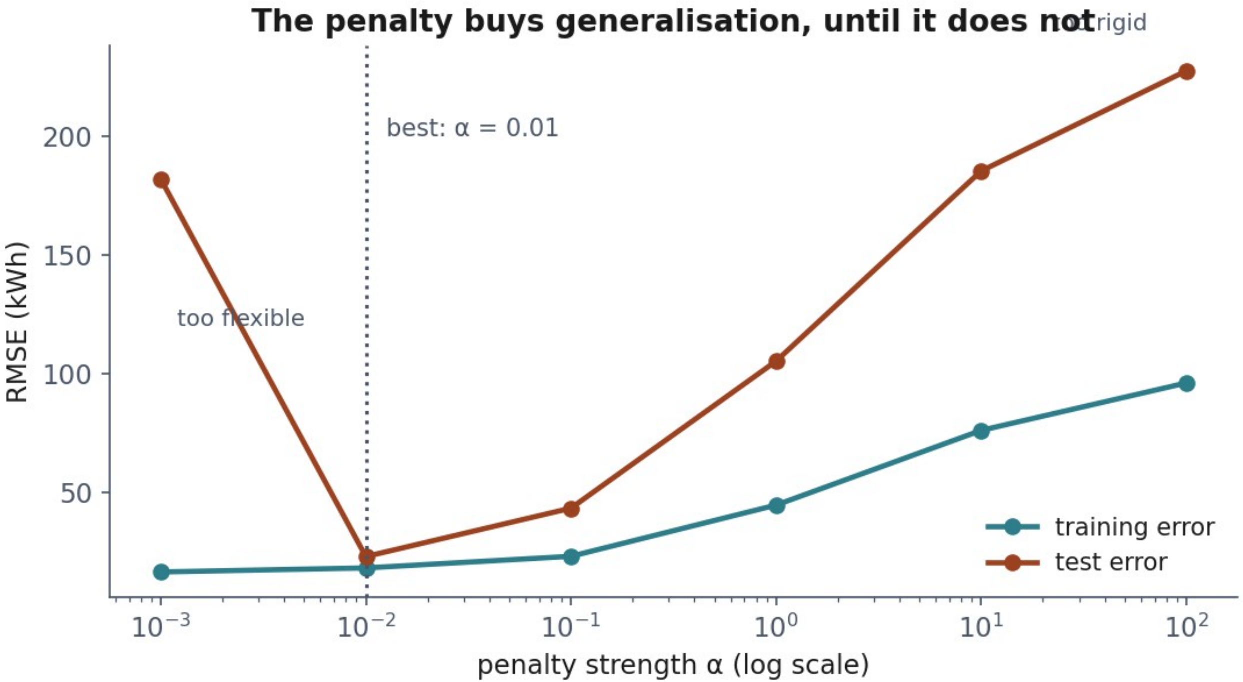
Two rules that are not optional

- The **intercept is never penalised** — it is not a claim about any feature
- Features **must be scaled first**, or the penalty is arbitrary

What a penalty buys

Model	Train	Test	max |w|
none	16.3	182.0	247,514
λ = 0.01	18.0	**22.8**	365
λ = 1	44.5	105.2	156
λ = 100	96.0	227.6	6

Too flexible, too rigid



Ridge has an exact solution too

$$\theta = (X^\top X + \lambda I)^{-1} X^\top y$$

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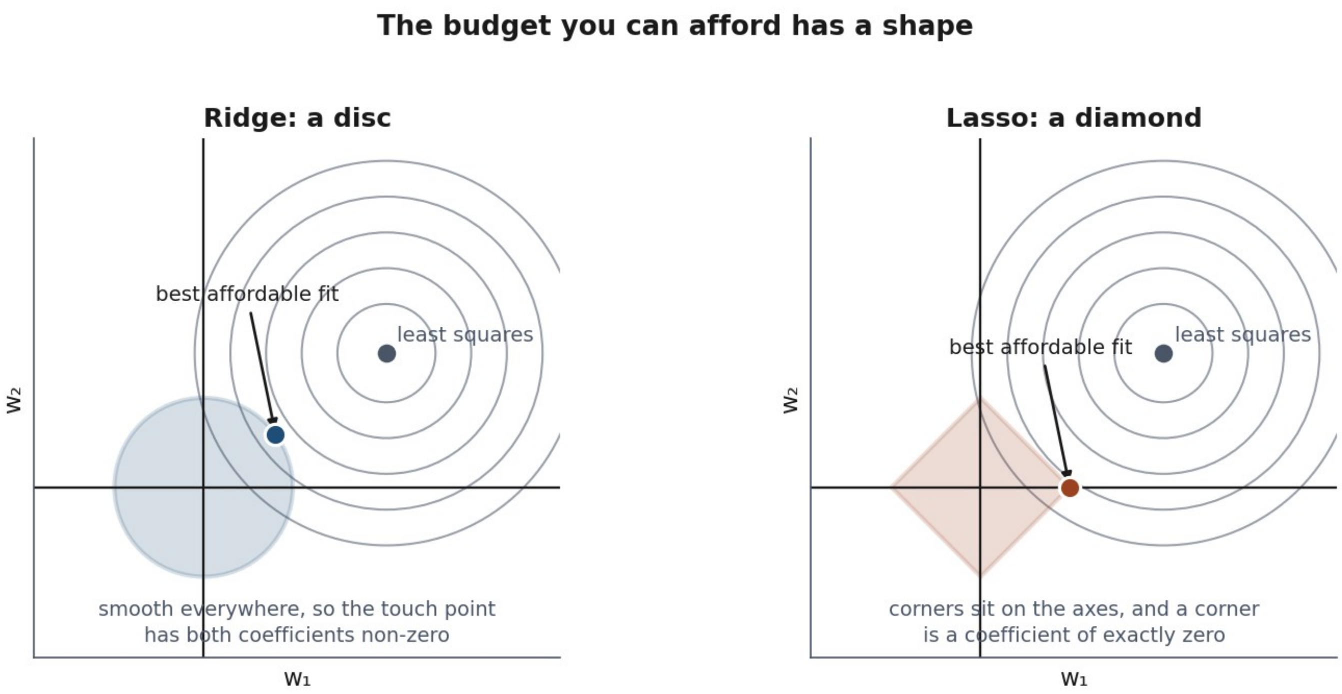
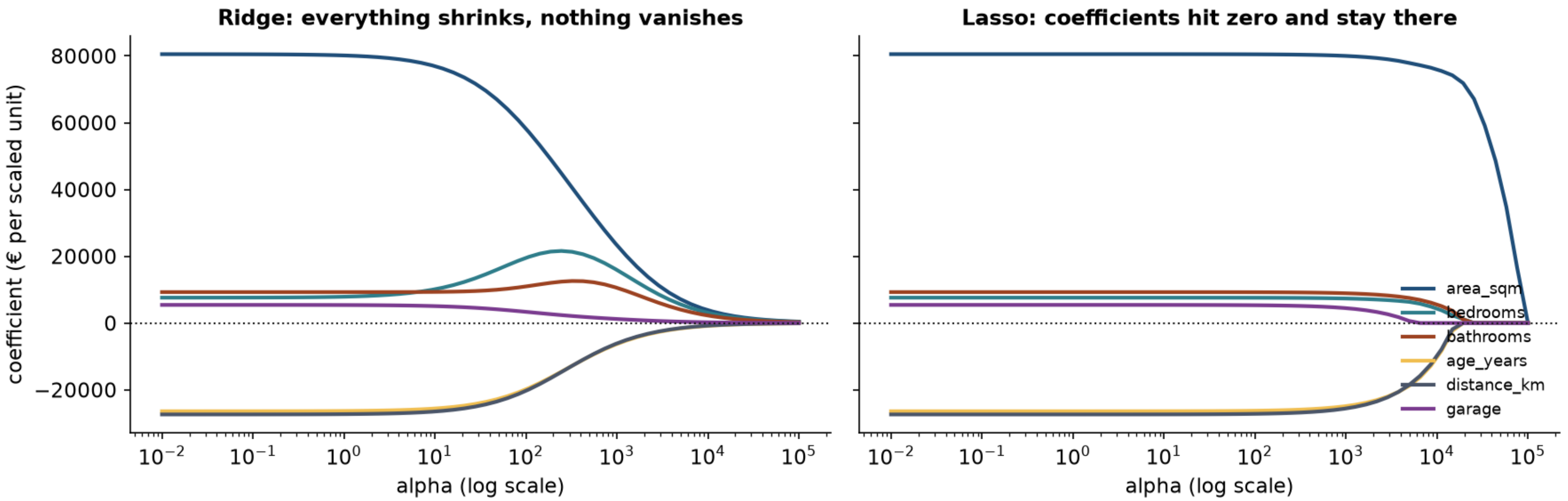
Why that one change fixes it

- Least squares fails when a direction costs nothing — the valley is flat
- The penalty makes **every** direction cost something
- Every eigenvalue shifts up by λ, so none of them can be zero

The picture behind Ridge vs Lasso



Ridge shrinks, Lasso selects



The order Lasso drops features

λ	Features kept
1,000	all six
10,000	five — `garage` dropped
20,000	three — `area`, `bedrooms`, `bathrooms`
40,000	one — `area_sqm`

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The case Ridge was invented for

`area_sqm` and `area_sqft` — the same measurement, two units.

Correlation: **0.999999**

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The coefficients are noise

Split	area_sqm	area_sqft
0	9,261	−640
1	14,535	−1,127
2	45,307	−3,989
3	39,061	−3,402

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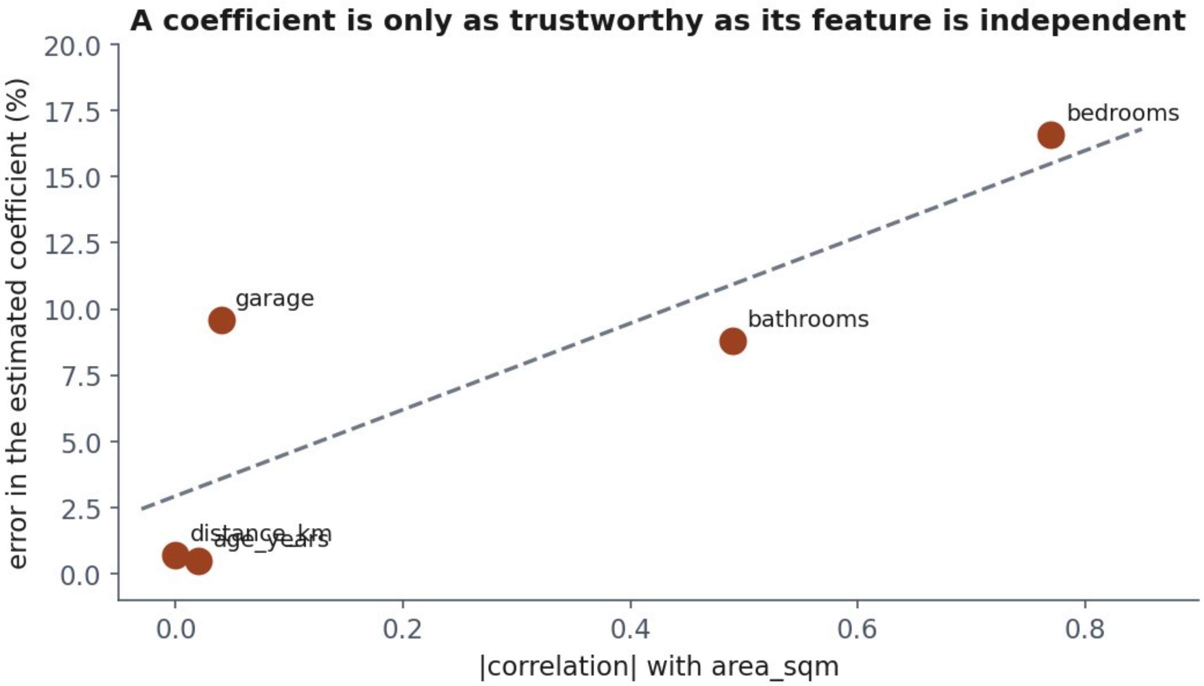
What a coefficient actually says

The change in the prediction for a one-unit change in that feature — **with every other feature held fixed**.

When can you trust a coefficient?



Notebook 3, live

Ridge on the disaster, the regularisation paths, the collinear case.

What we did today

- A model that claims a fixed amount per unit
- An exact solution — a projection — and when it fails
- An iterative one: walk downhill, mind your stride
- Flexibility helps until it does not
- Charging for size buys generalisation, and readable coefficients

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Homework — due Friday 16 October

`Exercises/03_regression.md`

Fit, regularise, and **explain which coefficients you believe**.